

HPHC Remote Patient Monitoring Platform — Detailed Design

Harbourfront Public Healthcare Cluster · Digital Health Platforms

Architecture, requirements, and validation criteria for build

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Clinical sponsor	Chair, HPHC Clinical Safety Board (CSB)
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Revision history

Version	Date	Changes
1.0	2026-08-21	First consolidated detailed design.
2.0	2026-09-26	FR-7 deduplication rule specified in full. NFR-9 policy basis restated. Section 7.3 and D-2: IoT Hub re-sized. Section 10.3 and D-10: Alert Dispatcher made active-active with externalised state, delivery assurance, and a pipeline heartbeat. Section 12.2 and D-14: outbound Observations posted as preliminary pending nurse validation. Section 15.2 and D-4: per-device X.509 identities. Section 22 acceptance criteria updated to match.

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1. Purpose and Scope

This document describes the architecture of the Harbourfront Public Healthcare Cluster (HPHC) Remote Patient Monitoring Platform (RPM-P) — a cluster-wide platform that receives continuous vital signs from wearable and bedside IoT devices, evaluates them against clinically governed rules and a machine-learning early-warning model, and routes alerts to the right clinician at the right time. It is written to a level of technical detail sufficient for an engineering team to begin implementation. Section 23 gives an explicit assessment of where that is and is not yet true.

RPM-P is not a replacement for bedside monitor alarms in high-acuity areas. It is a cluster-level platform that provides:

- Continuous monitoring of general-ward inpatients wearing biosensor patches, and of virtual-ward patients at home under the Mobile Inpatient Care@Home (MIC@Home) programme.
- Rule-based alerting (per-parameter thresholds and a partial NEWS2 score) and ML-based deterioration alerting.
- Alert routing, acknowledgement, and escalation to nurses and doctors via the cluster clinician mobile app and ward paging.
- Clinician dashboards: ward overview, patient trend view, and alert inbox.
- HL7 FHIR R4 integration with the cluster EMR for observations, ADT context, and manual observations.
- Device lifecycle management from receipt to decommissioning, including patient-device binding.
- Audit trails for every access to patient data, supporting PDPA and MOH data-governance obligations.

In scope

- General wards across the cluster's three acute hospitals (Harbourfront General Hospital, Telok Kurau Hospital, and Bukit Panjang Regional Hospital) and two community hospitals.
- MIC@Home virtual-ward patients managed by the cluster's virtual ward teams.
- Approximately 8,000 concurrently streaming devices at steady state: ~5,600 ward biosensor patches, ~1,400 bedside vital-signs monitors on step-down beds, and ~1,000 home monitoring kits.
- 24/7 operation.

Out of scope

- Intensive care units, high-dependency units, and operating theatres. These are served by vendor central stations with their own validated alarm systems; RPM-P does not ingest from them in this release (Pending Backlog B-1).
 - Patient-facing mobile application for home patients (Pending Backlog B-5). Home kits are pre-paired and require no patient app.
 - Any treatment-recommendation functionality. RPM-P raises alerts; it does not recommend treatment.
-

2. Requirements

Requirements are grouped into Functional Requirements (what the platform must do) and Non-Functional Requirements (the quality attributes it must exhibit). Each requirement carries an ID that is used again in Section 22 (Validation and Acceptance Criteria) and Section 23 (Build Readiness Assessment).

2.1 Functional Requirements

ID	Requirement
FR-1	The platform shall ingest vital signs from registered devices over MQTT 3.1.1 secured with TLS: heart rate, SpO2, respiratory rate, skin temperature, posture/activity, and (bedside monitors only) non-invasive blood pressure.
FR-2	Every telemetry message shall carry <code>device_id</code> , a per-device monotonically increasing sequence number, the device timestamp, and the gateway receive timestamp; the platform shall additionally record the IoT Hub enqueue timestamp.
FR-3	A device shall be associated with a patient only through a positive binding event that scans two identifiers (patient wristband or MIC@Home enrolment barcode, and device QR code). Telemetry from an unbound device shall never be attributed to a patient.
FR-4	The platform shall evaluate per-parameter threshold rules and a partial NEWS2 score (pNEWS2) for every bound patient, as defined in Section 4.
FR-5	The ML Early-Warning Service (EWS-ML) shall compute a deterioration risk score for every bound patient at least every 5 minutes and raise an alert when the score crosses the configured threshold for that patient's cohort.
FR-6	Every alert shall carry exactly one priority — High, Medium, or Low — assigned according to the mapping in Section 4.3.
FR-7	The Alert Service shall suppress an alert only when an alert for the same patient with the same <code>rule_id</code> (same parameter and rule) and the same or higher priority is still open, or was raised within the preceding 5 minutes. The 5-minute window is fixed from the first alert (it is not extended by suppressed alerts) and is measured on platform ingestion time, not device time. An alert of higher priority, or for a different parameter or rule, is never suppressed and is delivered immediately.
FR-8	Unacknowledged alerts shall escalate: High priority to the charge nurse at 60 s and to the ward doctor on call at 120 s; Medium priority to the charge nurse at 5 minutes. These escalation intervals are mandated by IEC 60601-1-8, clause 6.11 (distributed alarm systems).
FR-9	The platform shall raise technical alerts for device disconnection, sensor detachment, low battery, and poor signal quality, with the priorities given in Section 4.4.
FR-10	Clinicians shall be able to view a ward overview, a per-patient trend view, and an alert inbox, and shall acknowledge alerts under their own authenticated identity.
FR-11	The platform shall publish device-derived vital-sign observations to the EMR as HL7 FHIR R4 Observation resources, as specified in Section 12.
FR-12	The platform shall consume ADT events (admit, transfer, discharge) from the cluster integration engine to maintain care-team assignment and to unbind devices automatically on discharge.
FR-13	Every read of patient-identifiable data, every alert state change, every binding/unbinding event, and every configuration change shall be written to an immutable audit store recording actor, patient, action, timestamp, outcome, and reason (where applicable).
FR-14	Devices shall be managed through the lifecycle states defined in Section 14; no device may stream patient data unless it is in the Bound state.
FR-15	The platform shall produce anonymised extracts for model development and service evaluation, as specified in Section 17.4.
FR-16	Device firmware updates shall be cryptographically signed, deployed in stages, and applied only to devices not currently bound to a patient.

2.2 Non-Functional Requirements

ID	Requirement
NFR-1	The platform shall support 8,000 concurrently streaming devices at steady state, with capacity for a 25% surge (10,000 devices) during outbreak or surge-bed activation, without architectural change.
NFR-2	For High-priority physiological alerts, the time from sensor measurement to the alert being presented on the assigned clinician's device shall be ≤ 10 s at p95. For Medium priority, ≤ 30 s at p95.
NFR-3	The alert pipeline (ingestion $\rightarrow$ evaluation $\rightarrow$ notification) shall achieve $\geq 99.95\%$ monthly availability.
NFR-4	The alert pipeline shall remain available through the loss of a single availability zone with no alert loss, and through the loss of the entire cloud region with an RTO of ≤ 15 minutes.
NFR-5	The clinician dashboard shall reflect new readings within 5 s of ingestion and support 500 concurrent clinician sessions.
NFR-6	All transport shall use TLS 1.2 or higher. Devices shall authenticate individually to the platform; no anonymous device connections shall be accepted.
NFR-7	All PHI at rest shall be encrypted with customer-managed keys (CMK) held in Azure Key Vault Managed HSM, rotated at least annually, so that the cluster can revoke the platform's access to data independently of the cloud provider.
NFR-8	Audit records shall be immutable (write-once) and retained for the period set by the cluster records retention schedule.
NFR-9	All patient-identifiable data, including telemetry, backups, and logs, shall be stored and processed only within Singapore, as required by the cluster Cloud Data Residency Standard (HPHC-ISP-07, which implements the public-healthcare policy that sensitive patient data in commercial cloud remain onshore). Any access from outside Singapore (for example, vendor support) is additionally a transfer subject to the PDPA Transfer Limitation Obligation (Section 26) and requires contractual safeguards of a comparable standard of protection.
NFR-10	A new device model shall be onboarded by publishing a device adapter profile (parameter mapping, units, sampling rate) without code changes to the rule engine, Alert Service, or dashboards.
NFR-11	Clinical rule content (thresholds, pNEWS2 mapping, escalation timings) shall be versioned, CSB-approved before activation, and changeable without redeployment.
NFR-12	Every alert shall be traceable to the raw readings, rule or model version, and notification attempts that produced it.

3. Foundational Principles

These principles govern every design decision in the platform. Where a later section appears to conflict with one of these, the principle wins.

#	Principle
P1	Bedside alarms stay primary for inpatients. On wards with bedside monitors, the monitor's own audible alarm is the primary alarm. RPM-P is a secondary, remote notification layer. For MIC@Home patients, RPM-P is the primary monitoring channel and is designed accordingly.
P2	Clinicians own the thresholds. Every threshold, score mapping, and escalation timing is clinical content, approved by

#	Principle
	the CSB, versioned, and auditable. Engineers implement; they do not choose clinical values.
P3	The EMR is the legal medical record. RPM-P is a monitoring system, not a system of record. Anything that becomes part of the medical record lives in the EMR.
P4	Clinician validates before charting. Device-derived observations enter the EMR as preliminary data and become part of the legal medical record only after a nurse has validated them. Unvalidated device data is never presented as charted observation.
P5	No attribution without positive binding. A reading belongs to a patient only if a two-identifier binding event exists for that device and time.
P6	Care-team scoped access. Clinicians see patients they are caring for; anything else is break-glass and audited.
P7	Configuration, not code. New devices, new rules, and new wards are onboarded by configuration under change control.
P8	Minimum necessary data leaves the platform. Extracts, integrations, and logs carry only the fields their purpose needs.
P9	Every alert is explainable. From any alert, a clinician or auditor can trace back to the readings, rule or model version, and every delivery attempt.

4. Clinical Alerting Model

This section defines what the platform alerts on and at what priority. It was developed with the Clinical Informatics Office and approved by the CSB (decision CSB-2026-014). It is the clinical specification that Sections 8–10 implement.

4.1 Alert classes

Class	Source	Examples
Physiological — threshold	Rule engine (Section 8)	SpO2 sustained below threshold; HR sustained above threshold
Physiological — score	Rule engine (Section 8)	pNEWS2 aggregate or single-parameter red score
Physiological — predictive	EWS-ML (Section 9)	Deterioration risk above cohort threshold
Technical	Device and connectivity monitoring	Disconnected, sensor off, low battery, poor signal

4.2 Partial NEWS2 (pNEWS2)

NEWS2 (Royal College of Physicians, 2017) aggregates seven parameters: respiration rate, SpO2 (Scale 1 or Scale 2), supplemental oxygen, systolic blood pressure, pulse, level of consciousness (ACVPU), and temperature. Wearable patches measure only some of these continuously, so the platform computes a partial score with explicit rules rather than presenting a full NEWS2 it cannot measure.

NEWS2 parameter	Source in RPM-P	Freshness rule
Respiration rate	Patch (continuous)	Rolling 2-minute median

NEWS2 parameter	Source in RPM-P	Freshness rule
SpO2	Patch or bedside monitor (continuous)	Rolling 2-minute median; Scale 2 only if prescribed (see below)
Supplemental oxygen	Manual observation from EMR (FHIR)	Most recent within 4 h
Systolic BP	Bedside monitor NIBP or manual observation	Most recent within 4 h
Pulse	Patch or bedside monitor (continuous)	Rolling 2-minute median
Consciousness (ACVPU)	Manual observation from EMR (FHIR)	Most recent within 4 h
Temperature	Manual observation from EMR (FHIR)	Most recent within 4 h. Skin temperature from patches is not used for NEWS2 because it is not a validated surrogate for core temperature.

Rules:

- If every manually sourced parameter is within its freshness window, the platform computes the aggregate pNEWS2 and labels it "pNEWS2 (complete)".
- If any manually sourced parameter is stale, the platform does **not** compute an aggregate. It evaluates single-parameter scores only for the continuously measured parameters and raises a Low-priority "observations due" task for the ward nurse. The dashboard displays "pNEWS2 incomplete — manual observations due", never a number.
- SpO2 Scale 2 (target 88–92%, for patients with hypercapnic respiratory failure) is used only when a doctor has prescribed it in the EMR, as NEWS2 requires. The prescription is read through FHIR and has an expiry; on expiry the patient reverts to Scale 1.

4.3 Priority mapping

Platform priorities follow the alarm-priority semantics of IEC 60601-1-8: High requires immediate response, Medium requires prompt response, and Low requires awareness.

Condition	NEWS2 clinical risk	Platform priority
pNEWS2 (complete) aggregate ≥ 7	High	High
pNEWS2 (complete) aggregate 5–6	Medium	Medium
Any single continuous parameter scoring 3	Low–medium	Medium
Hard threshold breach (Table 4.3b)	—	High
EWS-ML risk above cohort threshold	—	Medium
pNEWS2 aggregate 1–4	Low	No alert; shown on dashboard

Table 4.3b — hard thresholds (CSB-2026-014, default adult values; per-patient overrides permitted by a doctor, with mandatory expiry ≤ 72 h):

Parameter	High-priority threshold	Persistence
SpO2 (Scale 1)	< 85%	10 s window
Heart rate	< 40 or > 140 bpm	10 s window
Respiratory rate	< 8 or > 30 /min	10 s window

The persistence window exists to suppress artefact (motion, probe displacement) and is deliberately part of the clinical definition of the alert condition.

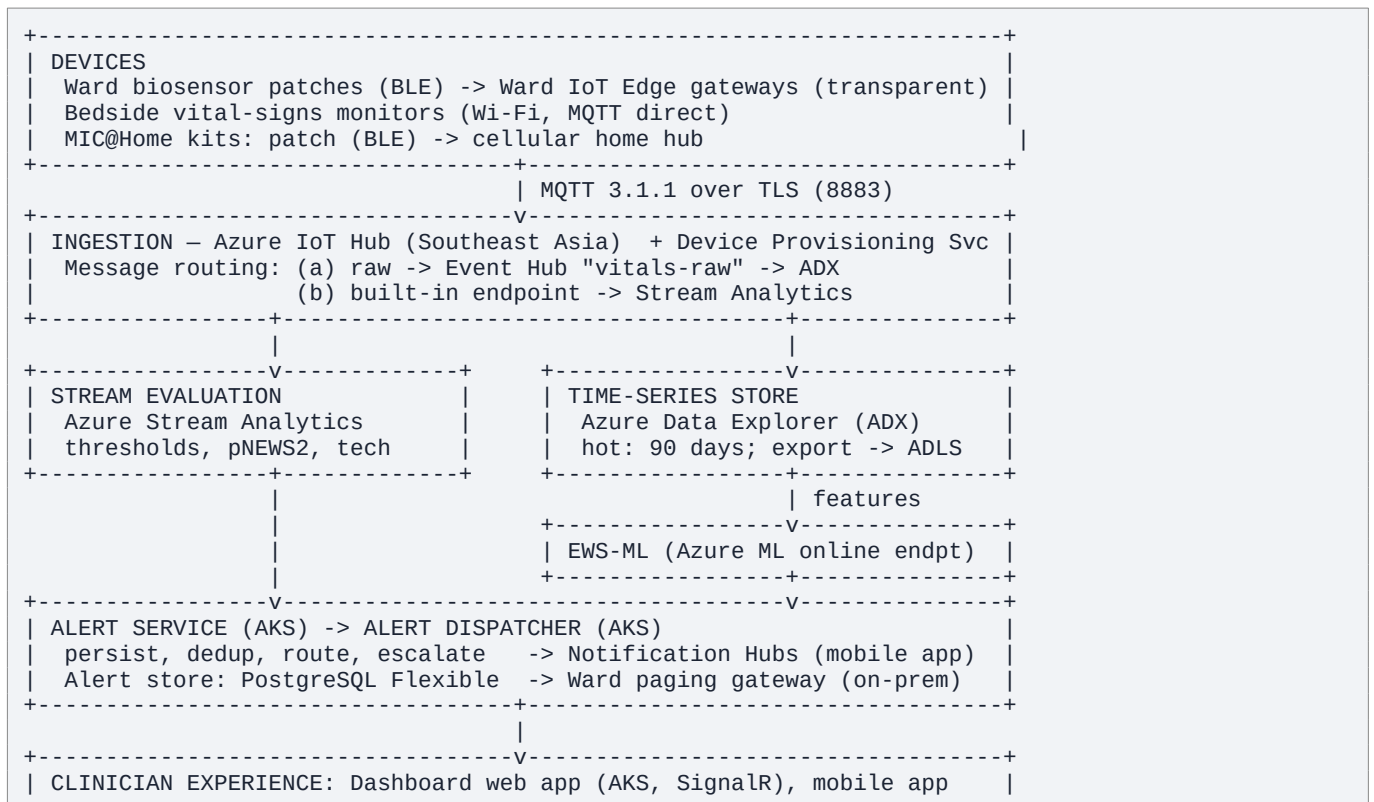
4.4 Technical alert priorities

Technical condition	Ward patient	MIC@Home patient
No data from bound device > 60 s	Low	Medium (after 5 min)
Sensor detached / no skin contact	Low	Medium (after 5 min)
Battery < 15%	Low	Low
Poor signal quality > 2 min	Low	Low

Technical alerts for MIC@Home are higher priority because there is no bedside alarm or nearby nurse to notice a disconnected patient.

5. Target Architecture

Every telemetry message, from every device, for every patient, flows through the same governed path. No clinician-facing component reads from a device directly.



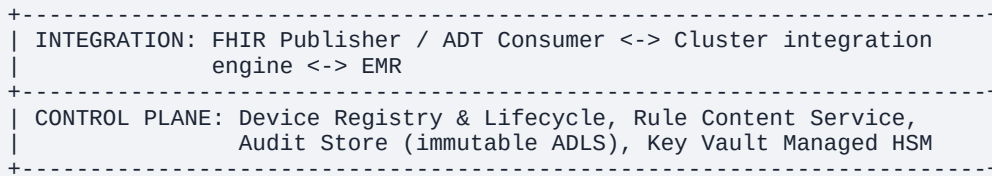


Figure 1 — Target architecture. Raw telemetry is persisted independently of evaluation, so a stream-job failure does not lose data.

Layer responsibilities

- **Devices and gateways** — measure, timestamp, buffer during link loss, and publish batched telemetry.
- **Ingestion** — device authentication, MQTT termination, message routing to storage and evaluation.
- **Stream evaluation** — windowed threshold rules, pNEWS2, technical-condition detection.
- **Time-series store** — raw and derived vitals for trend views, ML features, and FHIR publication.
- **EWS-ML** — periodic risk scoring per bound patient.
- **Alert Service and Dispatcher** — alert persistence, deduplication, routing to the responsible clinician, escalation, acknowledgement.
- **Clinician experience** — dashboard and mobile app.
- **Integration** — FHIR publication to the EMR; ADT and manual-observation consumption from the EMR.
- **Control plane** — device registry and lifecycle, CSB-approved rule content, audit, keys. The control plane is not in the real-time telemetry path.

6. Device Fleet and Message Envelope

6.1 Device classes

Class	Count (steady state)	Connectivity	Parameters	Publish interval
Ward biosensor patch	~5,600	BLE to ward IoT Edge gateway	HR, SpO2, RR, skin temp, posture	Batched, every 5 s
Bedside vital-signs monitor	~1,400	Hospital Wi-Fi, MQTT direct	HR, SpO2, RR, NIBP (on cycle)	Batched, every 5 s
MIC@Home kit (patch + hub)	~1,000	BLE to cellular home hub (4G)	HR, SpO2, RR, skin temp, posture	Batched, every 5 s

Each batched message carries the 1 Hz samples accumulated since the previous publish (five samples per parameter) and is ≤ 2 KB.

Buffering. Patches buffer up to 30 minutes of samples on-device when the BLE link to the gateway or home hub is lost, and replay them in order on reconnection. Home hubs buffer up to 4 hours when cellular connectivity is unavailable.

Clocks. Patches set their clock from the gateway or home hub at each BLE connection. The vendor specifies drift of ≤ 2 s per day between synchronisations. Ward gateways synchronise to the hospital NTP servers; home hubs use network-provided time (NITZ) with NTP fallback.

6.2 Message envelope

```

{
  "schema": "rpm.telemetry.v1",
  "device_id": "HGH-PATCH-004211",
  "model": "patch-vendorA-gen3",
  "fw": "3.4.1",

```

```

"seq": 1048213,
"device_ts": "2026-08-14T09:12:05.000+08:00",
"gw_id": "HGH-W12-GW02",
"gw_rx_ts": "2026-08-14T09:12:05.180+08:00",
"samples": {
  "hr": [{"t": 0, "v": 92}, {"t": 1, "v": 93}, ...],
  "spo2": [{"t": 0, "v": 95, "q": 0.98}, ...],
  "rr": [{"t": 0, "v": 18, "q": 0.91}, ...]
},
"battery_pct": 64,
"contact": true
}

```

Design notes:

- **seq** is a per-device monotonic counter persisted across reboots. Consumers use (device_id, seq) as the idempotency key, so replayed or duplicated publishes are discarded deterministically regardless of timestamps.
- **Three timestamps** are kept: device_ts (when measured), gw_rx_ts (when the gateway received it), and the IoT Hub enqueue time (added by the platform). The gap between them is recorded with every reading so clock problems and buffering are visible in the data rather than hidden.
- **Sample quality (q)** is passed through from the device's signal-quality index; readings with q below the device adapter profile's floor are stored but excluded from rule evaluation.
- **Patient identity is not in the envelope.** Devices do not know which patient they are on. Attribution happens in the platform, by joining on the binding table (Section 14.3). A device that is lost or reused never carries patient identifiers.
- **schema** is versioned; device adapter profiles (NFR-10) map vendor fields into this canonical envelope at the gateway.

7. Ingestion Layer

7.1 MQTT and topics

Devices and gateways connect to Azure IoT Hub using MQTT 3.1.1 over TLS 1.2 on port 8883. Ward patches connect through Azure IoT Edge gateways configured as transparent gateways, so each patch keeps its own device identity in IoT Hub and is individually addressable, revocable, and audited. Telemetry uses the standard IoT Hub device-to-cloud topic `devices/{device_id}/messages/events/` with message properties `schema`, `class`, and `site`.

7.2 Routing

Route	Condition	Endpoint	Purpose
raw-to-store	all telemetry	Event Hub vitals-raw → ADX streaming ingestion	Persist every accepted message
eval	all telemetry	Built-in endpoint (Event Hubs-compatible)	Stream Analytics input
lifecycle	class = lifecycle	Service Bus queue device-lifecycle	Battery, contact, firmware, error events

IoT Hub's built-in endpoint retains messages for 7 days, allowing replay into a rebuilt Stream Analytics job.

7.3 Capacity sizing

Metric	Value
Devices (steady state / surge)	8,000 / 10,000

Metric	Value
Message rate per device	1 per 5 s
Aggregate message rate (steady / surge)	1,600 / 2,000 msg/s
Daily messages (steady state)	$8,000 \times 17,280 = \sim 138 \text{ M/day}$
Message size	$\leq 2 \text{ KB}$ (one 4 KB metering unit)
IoT Hub tier	S3, 1 unit
Daily quota	300 M messages/day per S3 unit
Device-to-cloud send throttle	6,000 msg/s per S3 unit
Headroom over steady state	2.2× on daily quota; 3.75× on send rate
Headroom at surge (10,000 devices)	1.7× on daily quota ($\sim 173 \text{ M/day}$); 3× on send rate (2,000 msg/s)

Both the daily quota and the per-second send throttle were checked against the published IoT Hub quotas. At the S2 tier (6 M messages/day and 120 msg/s per unit) the fleet would need at least 24 units for the daily quota and 14 for the send rate, so S3 is both simpler and cheaper. A second S3 unit is added before the fleet exceeds 15,000 devices (one S3 unit saturates its daily quota at $\sim 17,000$ devices).

8. Stream Processing and Rule Evaluation

8.1 Job design

A single Azure Stream Analytics (ASA) job reads from the IoT Hub built-in endpoint and joins three reference inputs:

- **binding** — active device-to-patient bindings, refreshed from PostgreSQL every 60 s and on every binding change event.
- **rules** — the active, CSB-approved rule-content version (thresholds, overrides, pNEWS2 mapping).
- **manualobs** — latest manual observations per patient (BP, ACVPU, supplemental O2, temperature) fed by the FHIR consumer.

The job emits three outputs: physiological alert candidates and technical alert candidates to the Alert Service (via Service Bus topic alert-candidates), and pNEWS2 values to ADX for dashboard display.

8.2 Event-time configuration

Setting	Value	Rationale
Timestamp	TIMESTAMP BY device_ts	Rules must evaluate readings in the order they were measured, not the order they arrived
Out-of-order tolerance	5 s	Covers reordering across gateway and IoT Hub partitions
Late-arrival tolerance	5 s	Keeps window outputs timely for the latency target
Action for events outside	Drop	Avoids re-timestamping readings into the wrong window, which would

Setting	Value	Rationale
tolerance		distort persistence rules

8.3 Rule evaluation

- **Threshold rules** use a 10-second tumbling window per patient and parameter (the persistence window of Table 4.3b). A candidate is emitted when the window median crosses the threshold, and evaluation is on samples with quality at or above the profile floor.
- **pNEWS2** is evaluated every 30 s from 2-minute rolling medians of continuous parameters and the manual-observation reference input, applying the freshness rules of Section 4.2.
- **Technical conditions** use the absence of any message for a bound device for > 60 s (a left outer join of the binding reference against per-device message counts in a 60-second hopping window), contact flags, battery level, and quality indices.

Every candidate carries rule_id, rule-content version, the window boundaries, and the contributing sample sequence numbers (NFR-12).

9. ML Early-Warning Service (EWS-ML)

9.1 Model

Aspect	Specification
Task	Probability of clinical deterioration within 24 h (composite: rapid-response activation, unplanned ICU transfer, or death)
Algorithm	Gradient-boosted trees (LightGBM)
Features	1-, 4- and 12-hour summaries (median, slope, variability) of HR, SpO2, RR; most recent manual BP and ACVPU; age; Charlson comorbidity index from EMR problem list
Training data	HPHC general-ward inpatient admissions, 2021–2025 (~410,000 admissions)
Retrospective performance	AUROC 0.86 on a temporally held-out 2025 test set (inpatients)
Operating point	Cohort-specific threshold; inpatient threshold set at sensitivity 0.80, yielding 0.6 alerts per patient-day on the test set

9.2 Serving

EWS-ML runs as an Azure Machine Learning managed online endpoint (three instances across zones). A scheduler on AKS queries ADX feature views for every bound patient every 5 minutes and calls the endpoint in batches. Scores above the cohort threshold produce an alert candidate on the same Service Bus topic as rule-based candidates, carrying model name, version, and the top five feature contributions (SHAP) so the clinician can see why the alert was raised (P9).

9.3 Model governance

- Models are registered in the Azure ML registry with lineage to the training data snapshot and code commit.
- A model version becomes active only on CSB approval; activation is a rule-content change (NFR-11).
- Weekly monitoring reports alert rate per cohort, score distribution drift (population stability index), and outcome-linked performance once 24-hour outcomes are available.
- On drift beyond the agreed band, the Clinical Informatics Office reviews and may revert to the previous version.

9.4 Cohort use

For the inpatient cohort, EWS-ML alerts complement the rule-based alerts of Section 4. For the MIC@Home cohort, EWS-ML is the primary deterioration alert from Phase 4 (Decision D-15); rule-based hard thresholds remain active for MIC@Home as a safety net.

10. Alert Management and Notification

10.1 Alert Service

The Alert Service (AKS deployment, three replicas across zones) consumes alert-candidates, enriches each candidate with patient, location, and care-team context, persists it in the alert store (PostgreSQL), and passes it to the Alert Dispatcher. Persisting before dispatch means no candidate is acknowledged off the topic until it is durable.

10.2 Alert states

```
RAISED -> NOTIFIED -> ACKNOWLEDGED -> RESOLVED
      |               ^
      +--> ESCALATED -+
RAISED -> SUPPRESSED (duplicate, per FR-7)
```

All transitions are audited (FR-13), with actor, timestamp, and device used.

10.3 Alert Dispatcher

The Alert Dispatcher determines who should receive each alert and drives escalation.

- **Routing.** The responsible nurse is resolved from the ADT-driven care-team assignment (bed → nurse for the current shift). MIC@Home alerts route to the virtual-ward nurse on duty.
- **Channels.** The cluster clinician mobile app via Azure Notification Hubs (APNs/FCM), and the ward paging system via the on-premises paging gateway for High priority.
- **Deduplication.** Alerts are suppressed per FR-7 before notification. The deduplication key (patient_id, rule_id) and the open-alert window are held in zone-redundant Azure Cache for Redis, with PostgreSQL as the source of truth on cache loss.
- **Escalation.** Escalation timers per FR-8 start when the alert is NOTIFIED and are cancelled on acknowledgement. Each timer is a durable scheduled message on Service Bus (Premium, zone-redundant), so timers survive the loss of any Dispatcher instance; on firing, the Dispatcher re-reads the alert state from PostgreSQL and escalates only if the alert is still unacknowledged.
- **Deployment.** The Dispatcher runs active-active as three replicas across availability zones. Work is partitioned by ward using Service Bus sessions, so each ward's alerts are handled by one replica at a time and a failed replica's sessions are picked up by the survivors within the session lock timeout (30 s).
- **Delivery assurance.** Every notification attempt that is not confirmed delivered (Notification Hubs failure, APNs/FCM error, or no receipt from the paging gateway) is re-enqueued to the Service Bus queue alert-delivery and retried every 15 s until the alert is acknowledged. To guarantee that no alert is ever lost, messages on alert-delivery have no time-to-live and dead-lettering is disabled. The delivery worker processes each ward's session in strict FIFO order and does not advance past a message until its delivery is confirmed, so clinicians always receive alerts in the order they were raised.
- **Pipeline heartbeat.** A synthetic canary device per ward publishes a reading that triggers a dedicated heartbeat rule every 60 s. The Dispatcher records each heartbeat's end-to-end arrival; if no heartbeat has arrived for a ward for 90 s, every dashboard and mobile session for that ward shows a red "Monitoring pipeline interrupted — revert to manual observation" banner, and the platform on-call and ward charge nurse are paged through the paging gateway and by SMS.

10.4 Latency budget (NFR-2)

Segment	p95 budget
Sensor → device BLE notification to gateway / hub	0.5 s

Segment	p95 budget
Gateway or hub → IoT Hub (MQTT/TLS over hospital WAN or 4G)	0.7 s
IoT Hub → Stream Analytics input	1.0 s
Stream Analytics evaluation and output	1.0 s
Alert Service persist + Dispatcher routing	0.3 s
Notification Hubs → APNs/FCM → clinician device	1.5 s
Total	5.0 s

NFR-2 (≤ 10 s at p95) is therefore met with 5 s (50%) headroom.

10.5 Acknowledgement

Acknowledgement is performed in the mobile app or dashboard by the authenticated clinician, never by a shared account. Acknowledging stops escalation; resolving requires a reason code (treated, false alarm — artefact, false alarm — patient-specific, transferred). Reason codes feed the alert-fatigue review in Section 9.3 and Backlog B-8.

11. Clinician Dashboards

View	Content	Refresh
Ward overview	Bed grid with patient initials, latest HR/SpO2/RR, pNEWS2 or "incomplete", EWS-ML band, open alerts, device status	Push (SignalR), ≤ 5 s
Patient trend	1 h / 12 h / 72 h trends from ADX, alert markers, manual observations overlaid, binding history	On open, then push
Alert inbox	Open alerts for the clinician's patients, sorted by priority then age; acknowledge and resolve actions	Push
Virtual ward	MIC@Home patients with connectivity status, last reading age, and technical alerts	Push

The dashboard web app runs on AKS behind an in-region Azure Application Gateway (WAF v2) and is reachable only from the cluster network and the managed mobile app. Patient lists are scoped by care-team assignment (Section 16).

12. EMR Integration (HL7 FHIR R4)

12.1 Inbound

Data	Mechanism	Use
ADT (admit, transfer, discharge)	HL7 v2 ADT A01/A02/A03/A08 from the cluster integration engine, converted to FHIR Encounter	Care-team assignment, location, auto-unbind on discharge

Data	Mechanism	Use
Manual observations (BP, ACVPU, supplemental O2, temperature)	FHIR Subscription (rest-hook) on Observation, filtered to bound patients	pNEWS2 inputs
SpO2 Scale 2 prescription	FHIR ServiceRequest / flowsheet order, read on binding and on change	pNEWS2 scale selection
Problem list, age	FHIR Condition, Patient	EWS-ML features

12.2 Outbound

Every 15 minutes, the FHIR Publisher posts a transaction Bundle per bound patient containing one Observation per vital sign (the median of the interval), with `status = preliminary`, conforming to the FHIR R4 vital-signs profile. The EMR files these to a separate "device data — pending validation" flowsheet row. A nurse validates each interval in the EMR (accept, or reject with a reason such as artefact); accepted values are updated by the EMR to `final` and enter the charted record, and rejected values remain non-charted (P4). RPM-P never sets `final` itself.

Vital	LOINC	Unit (UCUM)
Heart rate	8867-4	/min
Oxygen saturation (pulse oximetry)	2708-6 + 59408-5	%
Respiratory rate	9279-1	/min
Blood pressure panel	85354-9 (8480-6, 8462-4)	mm[Hg]

Each Observation carries `device` (reference to the FHIR Device for the patch or monitor), `effectivePeriod` (the 15-minute interval), and a `meta.tag` identifying RPM-P as source. Skin temperature is not published, for the reason given in Section 4.2.

12.3 Error handling

Failed posts are retried with exponential backoff for up to 2 hours, then raised to the integration support queue. Because the time-series store is the platform's source for publication, a retry always posts the same medians for the same interval (idempotent on patient + code + effectivePeriod).

13. Data Model and Storage

13.1 Store summary

Data	Store	Retention	Encryption at rest
Raw and derived vitals	Azure Data Explorer, hot cache 30 days, retention 90 days	90 days hot, then export to ADLS	Microsoft-managed keys
Vitals archive	ADLS Gen2 (Parquet), cool tier	Per cluster records schedule	CMK (Managed HSM)
Alerts, bindings, device registry, rule content	Azure Database for PostgreSQL Flexible Server, zone-redundant HA	Alerts per records schedule	CMK (Managed HSM)

Data	Store	Retention	Encryption at rest
Audit	ADLS Gen2 immutable container (time-based retention policy, locked)	Per records schedule (NFR-8)	CMK (Managed HSM)
Model artefacts and features	Azure ML registry / ADX feature views	Per model lifecycle	Platform default

13.2 ADX vitals table

```
.create table Vitals (
  device_id: string,
  patient_id: string,           // joined at ingestion via binding update policy
  encounter_id: string,
  param: string,               // hr | spo2 | rr | skin_temp | nibp_sys | nibp_dia | posture
  value: real,
  unit: string,
  quality: real,
  seq: long,
  device_ts: datetime,
  gw_rx_ts: datetime,
  hub_enq_ts: datetime
)
```

An update policy joins incoming rows to the binding table on device_id and device_ts within the binding interval. Rows with no matching binding are written to vitalsunattributed with patient_id empty, for device-quality analysis only (P5).

13.3 PostgreSQL schema (core tables)

```
CREATE TABLE device (
  device_id      TEXT PRIMARY KEY,
  model          TEXT NOT NULL,
  class         TEXT NOT NULL,      -- patch | bedside | home_kit
  state         TEXT NOT NULL,      -- see Section 14.1
  fw_version     TEXT,
  site          TEXT,
  updated_at    TIMESTAMPTZ NOT NULL
);

CREATE TABLE device_binding (
  binding_id     UUID PRIMARY KEY,
  device_id     TEXT NOT NULL REFERENCES device(device_id),
  patient_id    TEXT NOT NULL,      -- cluster MRN
  encounter_id   TEXT NOT NULL,
  bound_at      TIMESTAMPTZ NOT NULL,
  unbound_at    TIMESTAMPTZ,
  bound_by      TEXT NOT NULL,      -- clinician user id
  unbind_reason TEXT,               -- discharge | manual | swap | fault
  EXCLUDE USING gist (device_id WITH =, tstzrange(bound_at, unbound_at) WITH &&)
);

CREATE TABLE alert (
  alert_id       UUID PRIMARY KEY,
  patient_id     TEXT NOT NULL,
  encounter_id   TEXT NOT NULL,
  class         TEXT NOT NULL,      -- threshold | score | predictive | technical
  rule_id       TEXT NOT NULL,
  content_ver    TEXT NOT NULL,      -- rule-content or model version
  priority      TEXT NOT NULL,      -- HIGH | MEDIUM | LOW
  state         TEXT NOT NULL,
  raised_at     TIMESTAMPTZ NOT NULL,
  evidence      JSONB NOT NULL      -- window bounds, seq list, SHAP top-5
);

CREATE TABLE alert_event (
  alert_id       UUID REFERENCES alert(alert_id),
  event          TEXT NOT NULL,      -- NOTIFIED | ESCALATED | ACK | RESOLVED | SUPPRESSED
  actor         TEXT,
  channel       TEXT,
  at            TIMESTAMPTZ NOT NULL,
  detail        JSONB
);
```

The exclusion constraint on `device_binding` makes it impossible for one device to be bound to two patients over overlapping intervals.

14. Device Lifecycle Management

14.1 Lifecycle states

```
RECEIVED -> REGISTERED -> AVAILABLE -> BOUND -> UNBOUND -> CLEANING -> AVAILABLE
                                     |           |
                                     |           +--> FAULT -> QUARANTINED -> AVAILABLE | DECOMMISSIONED
                                     +--> FW_UPDATING -> AVAILABLE
```

State	Entry condition	Streaming permitted?
RECEIVED	Goods receipt by Biomedical Engineering	No
REGISTERED	Identity created (Section 15.2), acceptance test passed	Test data only, to quarantine stream
AVAILABLE	Cleaned, charged, firmware current	No
BOUND	Two-identifier binding event (14.3)	Yes
UNBOUND	Discharge (ADT A03), manual unbind, or swap	No (data routed to unattributed table)
CLEANING	Returned to ward store	No
FAULT / QUARANTINED	Self-test failure, repeated quality failures, damage, or suspected tampering	No
FW_UPDATING	Staged firmware rollout (14.4)	No
DECOMMISSIONED	End of life, loss, or unrecoverable fault	No; identity disabled

14.2 Registration

Biomedical Engineering registers each device by scanning its QR code at goods receipt. Registration creates the device record (Section 13.3), the device identity (Section 15.2), and runs an acceptance test: connection, timestamp sanity (`device_ts` within 2 s of `gw_rx_ts` after sync), and a reference-signal check against a simulator.

14.3 Patient-device binding

- The nurse scans the patient wristband (inpatient) or MIC@Home enrolment barcode, then the device QR code, in the mobile app. Both identifiers are required (FR-3); manual typing of either is not permitted.
- The app displays the patient's full name, date of birth, and bed, and requires the nurse to confirm.
- The binding is written to PostgreSQL and pushed to the ASA binding reference input and the ADX binding table.
- On discharge (ADT A03), all bindings for the encounter are closed automatically. On transfer (A02), bindings persist and the care-team assignment follows the new location.
- A device swap (fault, battery) is a single transaction: unbind old, bind new, with the reason recorded.

14.4 Firmware updates

- Firmware images are signed by the vendor and countersigned by HPHC Biomedical Engineering; devices

- verify both signatures before applying (FR-16).
- Rollout uses IoT Hub automatic device management in stages: 5% of AVAILABLE devices, then 25%, then 100%, with automatic halt if the post-update self-test failure rate exceeds 1% at any stage.
- Updates are applied only to devices in AVAILABLE. A BOUND device is never updated mid-monitoring; it receives the update after it next passes through CLEANING.

14.5 Decommissioning

Decommissioning disables the device identity in DPS and IoT Hub, revokes its certificate, records the reason, and triggers a factory reset that wipes buffered samples. Lost devices are decommissioned immediately on report. Because the envelope never carries patient identifiers (Section 6.2), a lost device exposes no patient identity; buffered samples on it are unattributed physiological values.

15. Security Architecture

15.1 Network

- All PaaS services use private endpoints; public network access is disabled except IoT Hub, which accepts device connections from the internet for MIC@Home hubs over TLS 1.2 with IP filtering for ward gateways.
- AKS is a private cluster; egress goes through Azure Firewall with an allow-list.
- The on-premises paging gateway and integration engine are reached over ExpressRoute.
- The dashboard is exposed only through Application Gateway WAF v2 with cluster network and managed-device conditional access.

15.2 Device identity

Every device has its own X.509 identity. The private key is generated inside the device's secure element at the vendor factory and never leaves it. The vendor submits the resulting certificate signing requests to an HPHC device-issuing intermediate CA, whose key is held in Azure Key Vault Managed HSM; the signed certificate is written back to the device before shipment. DPS uses an X.509 CA-based group enrollment per device class anchored on that intermediate, so each device provisions with its own certificate and no shared secret exists on any device or in any firmware image. Revoking a single device means disabling its individual registration in DPS and its IoT Hub identity, which affects no other device. Ward IoT Edge gateways and home hubs use X.509 certificates issued by the same private CA hierarchy.

15.3 Data protection

- TLS 1.2+ everywhere (NFR-6); TLS 1.3 where the service supports it.
- Encryption at rest as in Section 13.1.
- Secrets and keys in Azure Key Vault Managed HSM; workload identities for AKS pods; no secrets in configuration files.

15.4 Threat model highlights

Threat	Control
Spooled clinician acknowledges alert	Entra ID with MFA; acknowledgement bound to authenticated user; audit
Compromised ward gateway	X.509 identity revocable per gateway; transparent gateway cannot alter per-device authentication
Tampering with rule content	Rule content changes require CSB approval workflow with two-person rule; versioned and audited
Replay of telemetry	(device_id, seq) idempotency
Data exfiltration from analytics	Private endpoints, egress allow-list, extract governance (Section 17.4)

Threat	Control

16. Identity, Access Control and Break-Glass

16.1 Authentication

Clinicians authenticate with the cluster Entra ID tenant using MFA. The mobile app runs only on cluster-managed devices (Intune), with app-level PIN or biometric re-authentication after 5 minutes of inactivity. Shared ward workstations use the cluster's existing tap-and-go badge session, so each action is attributable to an individual.

16.2 Authorisation

Role	Access
Ward nurse	Patients in care-team assignment for current shift; acknowledge and resolve alerts; bind and unbind devices
Charge nurse	All patients on the ward; escalation target
Doctor	Patients under their team; per-patient threshold overrides (with expiry)
Virtual-ward nurse / doctor	MIC@Home patients assigned to their virtual-ward team
Biomedical Engineering	Device registry and lifecycle; no patient identifiers
Clinical Informatics	Rule content authoring (activation requires CSB approval)
Platform operations	Infrastructure and logs; no patient identifiers in operational logs

Care-team assignment is derived from ADT location plus the shift roster, refreshed on every ADT event and at shift change.

16.3 Break-glass

A clinician who needs to view a patient outside their care-team scope may invoke break-glass by selecting a reason (emergency response, cross-cover, consult) and free text. Access is time-boxed to the remainder of the shift (maximum 12 hours), is shown with a visible banner, and generates an entry in the Data Protection Office review queue. Every break-glass event is reviewed retrospectively within 7 days; unjustified use is referred under the cluster's acceptable-use policy. Break-glass does not grant the ability to change thresholds or rule content.

16.4 Audit

Every patient-data read, alert action, binding change, override, and break-glass event is written to the immutable audit store (FR-13, NFR-8). Patients' data-access requests can be answered with a per-patient access report generated from the audit store.

17. Data Governance and Privacy

17.1 Legal and policy basis

- **PDPA 2012** — notification and consent for monitoring (obtained at MIC@Home enrolment and documented in the EMR; for inpatients, under the admission notice), purpose limitation, protection, retention limitation, access and correction, and the Transfer Limitation Obligation (Section 26) for any access to data from outside Singapore.
- **Cluster Cloud Data Residency Standard (HPHC-ISP-07)** — the source of the onshore storage and processing requirement in NFR-9.
- **Data breach notification** — PDPA Part 6A: notifiable breaches are reported to the PDPC as soon as practicable and no later than 3 calendar days after the cluster assesses a breach to be notifiable; affected individuals are notified where required.
- **MOH and cluster policy** — the cluster Data Governance Policy, Information Security Policy, and records retention schedule.

17.2 Purpose limitation

Purpose	Data	Access
Direct clinical care	Identified, full	Care team (Section 16)
Device and service operations	Device-level, no patient identifiers	Biomedical Engineering, platform operations
Model development and service evaluation	Anonymised extract (17.4)	Clinical Informatics, approved ML partners

17.3 Retention

Raw vitals are held in ADX for 90 days for trend review and model features, then exported to the ADLS archive and retained per the cluster records schedule. Charted observations live in the EMR (P3). Alerts and audit are retained per the records schedule.

17.4 Anonymised extracts

Extracts for model development and service evaluation are produced by a governed pipeline:

1. Direct identifiers are removed: NRIC/FIN, name, MRN, contact details, and free-text notes.
2. patient_id is replaced with a random study ID, unique per extract.
3. Retained fields: full vitals time series with timestamps to the second, ward and bed, age, sex, ethnicity, home postal code (MIC@Home cohort, used for the travel-time-to-hospital feature), diagnosis codes, and outcome labels.
4. The extract is written to the model-development workspace.

Because direct identifiers are removed, extracts are anonymised data and fall outside the PDPA's obligations. They may therefore be shared with approved ML partners in the partner's own development tenancy under the partner agreement, without further approval per extract.

18. Availability, Resilience and Disaster Recovery

18.1 Zone redundancy

All tiers are deployed across the three availability zones of the Azure Southeast Asia region:

Component	Zone posture
IoT Hub, DPS	Zone-redundant (platform-managed)

Component	Zone posture
Event Hubs, Service Bus (Premium)	Zone-redundant
Stream Analytics	Zone-redundant job placement
ADX	Zone-redundant cluster
PostgreSQL Flexible Server	Zone-redundant HA (synchronous standby in another zone)
AKS	Node pools across 3 zones; Alert Service 3 replicas
Azure ML endpoint	3 instances across zones

18.2 Region outage

The platform is deployed in a single region (Decision D-1). On a full regional outage:

- Inpatients: bedside monitor alarms remain primary (P1); wards revert to manual observation rounds per the cluster downtime procedure.
- MIC@Home: the virtual-ward team switches to scheduled telephone checks per the MIC@Home downtime procedure; patients follow the escalation instructions in their care plan.
- Platform recovery: redeploy from infrastructure-as-code and restore from zone-redundant backups when the region returns; estimated 4–8 hours after region availability.

18.3 Backups

PostgreSQL automated backups (35-day retention, zone-redundant storage); ADX follower and export to ADLS; infrastructure-as-code in the cluster repository. Backups remain in-region (NFR-9).

19. Prior Art and Reference Architecture

There is no single off-the-shelf platform that covers multi-vendor wearables, a cluster-wide alert routing model, and EMR integration under Singapore governance requirements. The closest references are standards and established patterns.

Reference	What it covers	Concept match	Gap
IHE Patient Care Device (PCD) — Device Enterprise Communication (DEC)	Transmission of device observations to enterprise systems	Observation payload, device identity, time semantics	HL7 v2-centric; we use FHIR R4 for outbound
IHE PCD — Alert Communication Management (ACM)	Alert dissemination to clinicians, status feedback	Alert lifecycle, delivery and acknowledgement status	Profile, not product; escalation logic is ours
IEEE 11073 SDC	Service-oriented device connectivity at point of care	Device semantic model	Acute-care device networks; wearables do not implement it
HL7 FHIR R4 vital-signs profile and Device resources	Observation and device representation	Outbound observations, device references	Does not address streaming or alerting
NEWS2 (RCP 2017)	Aggregate early-warning score	pNEWS2 design	Designed for intermittent manual observation, hence the partial-

Reference	What it covers	Concept match	Gap
			score rules in Section 4.2
Vendor central stations	Single-vendor monitoring and alarms in high-acuity areas	Alarm priority, central display	Single vendor; no wearable or home coverage; out of scope (B-1)

What we borrow

From	Borrowed for RPM-P
IHE PCD ACM	Alert states and delivery-status feedback model (Section 10.2)
IHE PCD DEC	Three-timestamp time semantics and device identity in observations
FHIR R4	Outbound Observation and Device representation
NEWS2	Score structure, Scale 2 rules, clinical-risk bands
IEC 60601-1-8	Alarm-priority semantics (High/Medium/Low)

What is specific to HPHC

- A single alert model spanning inpatient wards and home patients, with priorities adjusted for the absence of a bedside alarm at home.
- pNEWS2 with explicit freshness rules rather than imputing unmeasured parameters.
- Binding-based attribution so that devices never carry patient identity.

20. Confirmed Decisions

ID	Decision	Answer
D-1	Cloud and region	Azure Southeast Asia (Singapore), zone-redundant across 3 AZs; single region, no secondary region
D-2	Ingestion	Azure IoT Hub, S3 tier × 1 unit, with DPS (sized on daily quota and send throttle, Section 7.3)
D-3	Device protocol	MQTT 3.1.1 over TLS 1.2, port 8883
D-4	Device credentials	Per-device X.509 certificates, keys in secure element, issued by HPHC intermediate CA (Managed HSM); DPS CA-based group enrollment
D-5	Ward patch connectivity	BLE to IoT Edge transparent gateways
D-6	Stream evaluation	Azure Stream Analytics, event time = device_ts, out-of-order and late events outside tolerance dropped
D-7	Time-series store	Azure Data Explorer, 90-day retention, export to ADLS
D-8	Encryption at rest	ADX: Microsoft-managed keys (CMK deferred to reduce key-management overhead on the

ID	Decision	Answer
		ingestion cluster); PostgreSQL, ADLS, audit: CMK in Managed HSM
D-9	Alert store	PostgreSQL Flexible Server, zone-redundant HA
D-10	Alert Dispatcher topology	Active-active, 3 replicas across zones; dedup state in Redis, escalation timers as durable Service Bus scheduled messages; retry-until-acknowledged delivery queue; per-ward pipeline heartbeat
D-11	Deduplication rule	Same patient + same rule_id + same or higher priority, fixed 5-minute window on ingestion time (FR-7)
D-12	Escalation timings	High: 60 s / 120 s; Medium: 5 min (FR-8)
D-13	Notification channels	Clinician mobile app via Notification Hubs; on-prem paging gateway for High priority
D-14	Outbound EMR integration	FHIR R4 Observation Bundle every 15 min per bound patient, status preliminary; EMR sets final on nurse validation
D-15	EWS-ML for MIC@Home	EWS-ML replaces pNEWS2 aggregate alerting as the primary deterioration alert for the MIC@Home cohort from Phase 4; hard thresholds retained
D-16	pNEWS2	Partial score with freshness rules (Section 4.2), CSB-2026-014
D-17	Skin temperature	Displayed as trend only; not used in NEWS2 or published to EMR
D-18	Attribution	Binding table join; devices carry no patient identifiers
D-19	Research extracts	Anonymised per Section 17.4; shareable with approved ML partners
D-20	Firmware	Dual-signed, staged rollout, AVAILABLE devices only

21. Pending Backlog

Items confirmed as in scope but not yet fully designed:

ID	Item	Owner
B-1	Integration with vendor central stations in ICU/HDU (read-only alarm forwarding)	Clinical Informatics
B-2	Paging gateway API contract and delivery-receipt semantics with the paging vendor	Platform Integration
B-3	Regulatory determination for EWS-ML under HSA's framework for software as a medical device (classification and whether product registration or an exemption applies), and the prospective silent-mode validation protocol for the MIC@Home cohort	Clinical Safety Board /

ID	Item	Owner
		Regulatory Affairs
B-4	Alignment of outbound Observation profiles with the cluster FHIR implementation guide v2 when published	Integration Architecture
B-5	Patient-facing app for home patients (symptom entry, device status)	Digital Health Products
B-6	Archive tiering and cost optimisation for ADLS vitals archive	Platform Engineering
B-7	Multilingual patient onboarding materials for MIC@Home kits	Virtual Ward Programme
B-8	Alert-fatigue study design (post-go-live, using reason codes)	Nursing Informatics

22. Validation and Acceptance Criteria

Each requirement from Section 2 is validated by a specific method with a concrete pass/fail acceptance criterion. This table is the basis for the test plan in Phase 6 (Section 24).

22.1 Functional requirements

ID	Validation method	Acceptance criteria
FR-1	Device conformance test	Each device class publishes all listed parameters; each is ingested and visible in ADX with correct unit and value within the simulator's reference tolerance.
FR-2	Envelope contract test	Messages missing any of device_id, seq, device_ts, gw_rx_ts are rejected at the gateway adapter; stored rows carry all three timestamps.
FR-3	Binding safety test	Telemetry from an AVAILABLE (unbound) device appears only in VitalsUnattributed; a binding attempt with manual entry of either identifier is refused.
FR-4	Rule evaluation test	Simulator scripts for each Table 4.3b threshold and each pNEWS2 band produce exactly the expected candidates; stale manual observations produce "incomplete" and an observations-due task, never an aggregate.
FR-5	Scoring cadence test	Over a 2-hour run with 500 simulated patients, every bound patient receives a score at intervals ≤ 5 min.
FR-6	Priority mapping test	Every candidate type in Section 4.3 yields the specified priority.
FR-7	Deduplication test	(a) Two Medium SpO2 alerts for the same patient and rule 2 minutes apart: the second is SUPPRESSED. (b) A High SpO2 alert 3 minutes after an open Medium SpO2 alert is delivered. (c) A Medium HR alert 1 minute after an open Medium SpO2 alert is delivered. (d) A device with its clock offset by -10 minutes produces the same suppression outcomes as one with a correct clock.
FR-	Escalation	Unacknowledged High alert escalates to charge nurse at $60\text{ s} \pm 5\text{ s}$ and doctor at $120\text{ s} \pm 5\text{ s}$;

ID	Validation method	Acceptance criteria
8	timing test	acknowledgement before 60 s produces no escalation; killing the Dispatcher replica holding the ward session at 30 s does not delay escalation beyond 120 s $\pm$ 30 s.
FR-9	Technical alert test	Disconnect, contact loss, low battery, and low quality each produce the Section 4.4 priority for both cohorts.
FR-10	Dashboard functional test	Each view renders for a ward of 40 patients; acknowledgement records the authenticated user.
FR-11	FHIR contract test	Bundles validate against the FHIR R4 vital-signs profile with status preliminary and are accepted by the EMR test environment; values appear in the pending-validation row and become final only after nurse acceptance in the EMR.
FR-12	ADT simulation	A01/A02/A03/A08 messages produce the expected care-team changes; A03 closes all bindings for the encounter.
FR-13	Audit completeness test	A scripted session of reads, acknowledgements, binding changes, and a break-glass yields exactly one audit record per action with all fields populated.
FR-14	Lifecycle state test	A device in any state other than BOUND that publishes telemetry has its data routed to the unattributed table.
FR-15	Extract pipeline test	Extracts contain no field from the direct-identifier list in Section 17.4.
FR-16	Firmware test	An unsigned or singly-signed image is refused by the device; a BOUND device is not offered an update.

22.2 Non-functional requirements

ID	Validation method	Acceptance criteria
NFR-1	Load test	10,000 simulated devices at 1 msg/5 s for 24 h; no throttling errors, no message loss between IoT Hub and ADX.
NFR-2	Alert latency test	A test harness in Azure Southeast Asia publishes synthetic threshold-crossing messages directly to IoT Hub at 1,600 msg/s; time from IoT Hub enqueue timestamp to the alert row being committed in the alert store is $\leq$ 10 s at p95 (High) and $\leq$ 30 s at p95 (Medium).
NFR-3	Availability monitoring	Monthly availability computed from the per-ward pipeline heartbeat $\geq$ 99.95% over the first three production months; stopping the canary for a ward raises the interruption banner within 90 s.
NFR-4	Zone failure drill	With one zone's resources made unavailable, alert candidates continue to be produced and delivered; no alert row lost.
NFR-5	Dashboard performance test	500 concurrent sessions; new reading visible $\leq$ 5 s after ingestion at p95.
NFR-6	Transport security scan	No endpoint accepts TLS $<$ 1.2; a device connection without valid credentials is refused.

ID	Validation method	Acceptance criteria
NFR-7	Key management review	Every store in Section 13.1 marked CMK shows a Managed HSM key; rotation date within 12 months.
NFR-8	Immutability test	An attempt to modify or delete an audit blob within the retention period fails.
NFR-9	Residency audit	All resources and backups are deployed in Southeast Asia; no geo-redundant storage is enabled.
NFR-10	Onboarding dry run	A new patch model is onboarded by publishing an adapter profile with zero code changes in rule engine, Alert Service, or dashboards.
NFR-11	Rule-content change test	A threshold change takes effect without redeployment only after CSB approval is recorded; unapproved content cannot be activated.
NFR-12	Traceability test	For 100 random alerts, the evidence field resolves to stored raw readings and a rule or model version, and every delivery attempt is in alert_event.

23. Build Readiness Assessment

This section answers whether an engineering team could build each component from this document without further design decisions.

Component	Ready?	What remains to decide
Device envelope and adapter profiles	Ready	Envelope and design notes are concrete (Section 6.2).
Ingestion (IoT Hub, DPS, routing)	Ready	Tier, routes, and identity model specified (Sections 7, 15.2).
Stream evaluation (ASA)	Ready	Inputs, event-time settings, and rules specified (Section 8).
Clinical alerting model	Ready	CSB-approved (Section 4).
EWS-ML serving	Ready	Serving pattern specified; model trained (Section 9).
Alert Service and Dispatcher	Ready	Active-active topology, durable timers, delivery assurance, and heartbeat specified (Section 10.3). Paging delivery receipts depend on B-2.
Dashboards	Mostly ready	Views specified; detailed UX to be produced with Nursing Informatics.
FHIR integration	Mostly ready	Outbound and inbound specified; profile alignment pending (B-4).
Device lifecycle	Ready	States, binding, firmware, decommissioning specified (Section 14).
Security and access	Ready	Sections 15–16.

Component	Ready?	What remains to decide
Data governance and extracts	Ready	Section 17.
DR	Ready	Single-region posture with downtime procedures (Section 18).

Overall conclusion. The platform is buildable from this document for Phases 1–5. The remaining items in Section 21 are either out of the initial scope (B-1, B-5, B-7), integration details that do not change the architecture (B-2, B-4, B-6), or post-go-live activities (B-8). B-3 runs in parallel with Phase 4 build.

24. Build Phases

#	Phase	Scope	Depends on open items?
1	Foundation	Landing zone, networking, IoT Hub, DPS, ADX, PostgreSQL, Key Vault, audit store	No — ready
2	Devices and lifecycle	Adapter profiles, IoT Edge gateways, registration, binding app, firmware pipeline	No — ready
3	Rules, alerts, dashboard — pilot	ASA job, Alert Service and Dispatcher, mobile app notifications, dashboards; pilot on 2 wards at Harbourfront General Hospital	No — ready
4	EWS-ML and EMR integration	EWS-ML serving; MIC@Home go-live with EWS-ML as primary deterioration alert; FHIR outbound and inbound	No — ready
5	Cluster scale-out	All in-scope wards and full MIC@Home cohort; paging integration	Partially — B-2
6	Validation and hardening	Section 22 test plan, zone failure drills, security testing, clinical safety case finalisation	Follows Section 22

This document, together with the companion Conceptual Design and Clinical Safety Case, represents the design phase in full. No implementation has begun.